

GOOGLE CYBERSECURITY CERTIFICATE

Network Hardening

Google Cybersecurity Professional Certificate · Study Guide

A survey of the core network hardening tasks used to reduce attack surface — baseline configuration management, firewall and patch maintenance, MFA, port filtering, access privilege management, and NIST-aligned password policy.

Track	Google Cybersecurity Professional Certificate
Course	Assets, Threats, and Vulnerabilities
Focus	Attack surface reduction
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1 Hardening Tasks

Network hardening reduces the number of ways an attacker can get in by tightening configuration, limiting exposure, and keeping systems current. Each task below targets a different layer of the network.

Task	Purpose
Baseline configuration management	Maintain a documented, approved configuration standard so deviations (misconfigurations, unauthorized changes) are easy to spot
Firewall maintenance	Keep rules current with the latest traffic standards and threat intelligence
Patch management	Apply vendor security patches promptly to close known vulnerabilities
MFA implementation	Require a second verification factor for access, independent of password strength
Port filtering / disabling unused ports	Close every port that has no active business justification
Network access privilege management	Grant only the access level a role actually needs
Password policies (NIST guidelines)	Enforce length, complexity, and rotation rules aligned to current NIST guidance

NOTE None of these controls is a silver bullet on its own — hardening is cumulative. A patched, firewalled host with an open unused port and a weak password policy is still meaningfully exposed.

2 Quick Reference



Network hardening — quick reference

Hardening checklist:

- ☐ Baseline configuration documented and enforced
- ☐ Firewall rules reviewed on a schedule, not just after incidents
- ☐ Patches applied promptly (track a target SLA per severity)
- ☐ MFA required for all privileged and remote access
- ☐ Unused ports/services disabled
- ☐ Access reviewed against least privilege
- ☐ Password policy matches current NIST guidance